

Mental Math — Level Definitions (v1)

1 Overview

Each domain (Addition, Subtraction, Multiplication, Division) is divided into ten progressive levels. A **pattern** defines the shape of questions generated for that level: the number of digits in each operand, the required number of carries or borrows, and the cutoff time within which a correct answer must be submitted to count as fast.

Levels are designed to be completed in order within a domain. Progression to the next level requires demonstrating consistent performance on the current one (see `progression-algo/algorithm.tex`).

Column definitions used in the tables below:

- **Level** — sequential difficulty step within the domain (1 = easiest).
- **Description** — plain-language summary of the pattern.
- **Operands** — digit widths of the left and right operands (or dividend / divisor for division).
- **Carries / Borrows** — exact number of carry or borrow operations required by the generated question.
- **Cutoff (ms)** — time budget per question; answers submitted within this window are considered fast.

2 Addition (ADD)

Level	Description	Operands	Carries	Cutoff (ms)
1	1-digit + 1-digit, no carry	1 + 1	0	4 000
2	1-digit + 1-digit, with carry	1 + 1	1	4 000
3	1-digit + 2-digit, no carry	1 + 2	0	4 500
4	1-digit + 2-digit, with carry	1 + 2	1	4 500
5	2-digit + 2-digit, no carry	2 + 2	0	5 000
6	2-digit + 2-digit, one carry	2 + 2	1	5 000
7	2-digit + 2-digit, two carries	2 + 2	2	5 500
8	2-digit + 3-digit, no carry	2 + 3	0	5 500
9	2-digit + 3-digit, one carry	2 + 3	1	6 000
10	3-digit + 3-digit, no carry	3 + 3	0	6 000

Notes. Carry count specifies exactly how many column additions overflow into the next column. Level 1 guarantees single-digit sums (e.g. $3 + 4 = 7$); Level 5 requires handling two-digit arithmetic with no carrying; Levels 6–10 extend to two- and three-digit operands with increasing carry requirements.

3 Subtraction (SUB)

Level	Description	Operands	Borrows	Cutoff (ms)
1	1-digit – 1-digit, non-negative	$1 - 1$	0	4 000
2	2-digit – 1-digit, no borrow	$2 - 1$	0	4 500
3	2-digit – 1-digit, with borrow	$2 - 1$	1	4 500
4	2-digit – 2-digit, no borrow	$2 - 2$	0	5 000
5	2-digit – 2-digit, with borrow	$2 - 2$	1	5 000
6	3-digit – 1-digit, no borrow	$3 - 1$	0	5 000
7	3-digit – 1-digit, with borrow	$3 - 1$	1	5 500
8	3-digit – 2-digit, no borrow	$3 - 2$	0	5 500
9	3-digit – 2-digit, with borrow	$3 - 2$	1	6 000
10	3-digit – 3-digit, no borrow	$3 - 3$	0	6 000

Notes. Negative results are never generated (`allowNegative: false` for all levels). Level 1 constrains the generator so that the right operand is always $\leq$ the left operand, avoiding borrows by construction. Levels 6–10 introduce three-digit minuends, progressing from simple subtraction up to full three-digit minus three-digit.

4 Multiplication (MUL)

Level	Description	Operands	Carries	Cutoff (ms)
1	1-digit $\times$ 1-digit, no carry	1×1	0	4 000
2	1-digit $\times$ 1-digit, with carry	1×1	1	4 000
3	1-digit $\times$ 2-digit, no carry	1×2	0	4 500
4	1-digit $\times$ 2-digit, one carry	1×2	1	4 500
5	1-digit $\times$ 2-digit, two carries	1×2	2	5 000
6	1-digit $\times$ 3-digit, no carry	1×3	0	5 000
7	1-digit $\times$ 3-digit, one carry	1×3	1	5 500
8	1-digit $\times$ 3-digit, two carries	1×3	2	5 500
9	1-digit $\times$ 3-digit, three carries	1×3	3	6 000
10	2-digit $\times$ 2-digit, no carry	2×2	0	6 000

Notes. For multiplication, a carry occurs when a partial product exceeds 9 and must be added to the next column. Levels 6–9 extend to a three-digit multiplicand, with the carry count rising from zero to three. Level 10 introduces two-digit multipliers for the first time.

5 Division (DIV)

All division questions are **exact integer** divisions (no remainder). Difficulty is controlled by constraining the number of digits in the operands and the length (number of digits) of the quotient.

Level	Description	Dividend / Divisor	Quotient Digits	Cutoff (ms)
1	1-digit $\div$ 1-digit	$1 \div 1$	1	4 000
2	2-digit $\div$ 1-digit	$2 \div 1$	1	4 500

Level	Description	Dividend / Divisor	Quotient Digits	Cutoff (ms)
3	2-digit $\div$ 2-digit	$2 \div 2$	1	5 000
4	2-digit $\div$ 1-digit	$2 \div 1$	2	5 500
5	3-digit $\div$ 2-digit	$3 \div 2$	1	6 000
6	3-digit $\div$ 1-digit	$3 \div 1$	2	6 500
7	3-digit $\div$ 2-digit	$3 \div 2$	2	6 500
8	3-digit $\div$ 1-digit	$3 \div 1$	3	7 000
9	4-digit $\div$ 2-digit	$4 \div 2$	2	7 500
10	4-digit $\div$ 3-digit	$4 \div 3$	2	8 000

Notes. Questions are generated by first choosing a valid divisor and a quotient with the targeted number of digits, then computing the dividend as their product. This guarantees an integer result and controls the exact structural shape of the mental operation. Levels 6–10 scale up to four-digit dividends with increasingly large divisors.

6 Summary: Cutoff Times

The tables below show the cutoff time (in milliseconds) for each level across all four domains. **Levels 1–5**

Domain	L1	L2	L3	L4	L5
Addition	4 000	4 000	4 500	4 500	5 000
Subtraction	4 000	4 500	4 500	5 000	5 000
Multiplication	4 000	4 000	4 500	4 500	5 000
Division	4 000	4 500	5 000	5 500	6 000

Levels 6–10

Domain	L6	L7	L8	L9	L10
Addition	5 000	5 500	5 500	6 000	6 000
Subtraction	5 000	5 500	5 500	6 000	6 000
Multiplication	5 000	5 500	5 500	6 000	6 000
Division	6 500	6 500	7 000	7 500	8 000

Division carries the longest cutoff times at higher levels because reasoning through multi-digit division requires more working-memory steps than the equivalent addition or multiplication. Addition, Subtraction, and Multiplication share the same cutoff schedule across Levels 6–10.